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ELECTRIC GRID NETWORK OPTIMIZATION USING MINIMUM SPANNING TREE ALGORITHMS

A CAPSTONE PROJECT REPORT

*A Capstone Project Report submitted in partial fulfilment of the requirements
for the Course of*

CSA0609 – DESIGN AND ANALYSIS OF ALGORITHM

to the award of the degree of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**ELECTRIC GRID NETWORK OPTIMIZATION USING MINIMUM SPANNING TREE ALGORITHMS**" has been carried out by **Bhuvanesh. S (Reg. No. 192521004)** under the supervision of **Dr. P. Solainayagi** and is submitted in partial fulfilment of the requirements for the current semester of the **BACHELOR OF TECHNOLOGY IN IT** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, **Bhuvanesh. S (192521004)** of the Department of Information Technology, SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled " **ELECTRIC GRID NETWORK OPTIMIZATION USING MINIMUM SPANNING TREE ALGORITHMS** " is the result of my own bonafide efforts. To the best of my knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. I further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. I confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

Place: Chennai

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Signature of the Student with Name

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INDIVIDUAL CONTRIBUTION STATEMENT

(Mandatory for all team projects)

Student / Register No.	Specific Responsibilities	Design & Development	Testing & Analysis	Report Contribution	Approx. (%)
192521004- BHUVANESH S	Analyzed the electric grid network problem and studied Minimum Spanning Tree (MST) concepts.	Designed the graph representation and implemented Kruskal's Algorithm using edge sorting and Union-Find.	Tested Kruskal's Algorithm with different grid inputs and verified minimum cost and connectivity.	Contributed to Introduction, Problem Statement, Objectives, MST Concepts, and Kruskal's Algorithm.	35%
192425311 - ATHIVARAM JAYAPRAKASH	Studied graph algorithms and analyzed the requirements for optimizing electrical connections.	Studied graph algorithms and analyzed the requirements for optimizing electrical connections.	Studied graph algorithms and analyzed the requirements for optimizing electrical connections.	Studied graph algorithms and analyzed the requirements for optimizing electrical connections.	33%
192421412 - KRISHNAPRASATH E	Worked on system integration, input/output handling, and performance evaluation.	Worked on system integration, input/output handling, and performance evaluation.	Performed overall testing, verified MST outputs, and analyzed algorithm performance and efficiency.	Worked on system integration, input/output handling, and performance evaluation.	32%
Total					100%

ABSTRACT

Electric power transmission and distribution networks contain substations, generating stations, and candidate transmission corridors that must be connected at acceptable cost while maintaining network reachability. A planning model that selects every available corridor can produce unnecessary capital expenditure, whereas a model that selects too few corridors can disconnect supply points. This project formulates the base network-design problem as a weighted undirected graph and applies Minimum Spanning Tree (MST) algorithms to obtain a connected subnetwork of minimum total edge weight.

The edge weight represents a normalized installation score combining route length, indicative construction cost, and loss-related penalty; the score is used for algorithmic study and is not a protection or operational setting. Kruskal's algorithm with disjoint-set union is implemented as the primary method, and Prim's algorithm is implemented as an independent comparison. A reproducible eight-substation synthetic network is used to verify feasibility, acyclicity, edge-count optimality, and total weight.

Both algorithms produce the same spanning-tree cost of 26 units for the illustrative network, using seven of the seventeen candidate corridors. The selected corridors are S4–S5, S2–S3, S6–S7, S1–S2, S5–S6, S3–S4, and S7–S8. Benchmarking on generated connected graphs indicates the expected practical behaviour of edge-sorting Kruskal and adjacency-based Prim for small planning instances

Keywords: Minimum Spanning Tree; Kruskal's Algorithm; Prim's Algorithm; Electric Grid; Graph Optimization; Network Planning

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit
MST	Minimum Spanning Tree	—
DSU	Disjoint-Set Union	—
V	Set of vertices/substations	count
E	Set of candidate edges/corridors	count
w(e)	Weight of edge e	normalized unit
T	Selected spanning tree	—
O($\cdot$)	Asymptotic complexity	—

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

An electric grid is a network in which generating stations, substations, and loads are connected through transmission and distribution corridors. During preliminary planning, engineers evaluate alternative corridors under competing objectives such as connectivity, capital cost, geographical distance, losses, constructability, and future expansion. Graph theory offers a compact abstraction: substations are represented by vertices, candidate corridors by edges, and a numerical weight expresses the relative cost of using each corridor.

A spanning tree connects all vertices without cycles. A Minimum Spanning Tree is a spanning tree whose sum of edge weights is minimum. Kruskal's algorithm constructs the tree by considering edges in nondecreasing weight order and rejecting any edge that creates a cycle. Prim's algorithm grows a tree from a starting vertex by repeatedly adding the least-weight edge crossing the current cut [1] [2].

1.2 Need for the Project

Grid corridors are expensive, geographically constrained assets. A transparent algorithmic baseline is useful before more complex reliability and power-flow models are applied. The need is significant for planners and students because it demonstrates how a discrete optimization method converts a network-design objective into a verifiable set of connections. Existing informal selection methods may overlook cycles, disconnected substations, or the cumulative effect of many locally attractive choices.

- The problem affects utilities, planners, communities, and customers because corridor selection influences cost, land use, maintainability, and service reachability.
- The engineering challenge is to minimize a weighted connection cost while preserving connectivity and avoiding redundant cycles in the base design.
- A mathematically verifiable solution is valuable for education and can serve as a preprocessing stage for capacity, reliability, and power-flow studies.

1.3 Problem Statement

Given a connected undirected graph $G=(V,E)$ representing substations and candidate corridors, select a subset $T \subseteq E$ that connects every substation, contains exactly $|V|-1$ edges, has no cycle, and minimizes $\sum w(e)$ for $e \in T$, subject to the assumptions that weights are non-negative, candidate

corridors are feasible, and operational constraints such as power-flow limits are outside the base model.

1.4 Project Aim

To design, implement, and evaluate a Minimum Spanning Tree based optimization workflow for selecting a minimum-weight connected electric-grid network from candidate corridors.

1.5 Project Objectives

- Model an illustrative electric grid as a weighted graph with substations and candidate corridors.
- Implement Kruskal’s algorithm using disjoint-set union with path compression and union by rank.
- Implement Prim’s algorithm as an independent cross-check of the selected total weight.
- Verify connectivity, acyclicity, edge count, and objective value on a reproducible dataset.
- Benchmark execution time on generated connected graphs and interpret the engineering implications.
- Document limitations, risks, ethical considerations, and extensions required for real deployment.

1.6 Scope

Included are graph modelling, edge-weight definition for a normalized planning score, Kruskal and Prim implementations, deterministic testing, illustrative visualisation, and complexity analysis. Excluded are AC/DC power-flow calculations, transient stability, short-circuit protection coordination, N–1 contingency compliance, detailed route surveying, land acquisition, and regulatory approval. The boundary condition is that the input graph is connected; if it is disconnected, the implementation reports that a spanning tree cannot be formed.

1.7 Expected Engineering Outcome

The outcome is a reproducible algorithmic prototype and decision-support workflow that returns the selected corridors, total normalized cost, feasibility checks, and benchmark evidence. It is an educational optimization model rather than an operational grid-control system.

Notation	Meaning	Assumption
V	Substations	All nodes must be connected
E	Candidate corridors	Undirected and feasible
w(e)	Normalized corridor score	Non-negative
T	Selected network	Tree with $ V -1$ edges

CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

The review focused on foundational graph-algorithm texts, original algorithm publications, power-system planning references, and publicly available standards and technical guidance. Sources were selected for authority, direct relevance to spanning trees, and clarity about the distinction between a connectivity-cost model and a complete power-system planning model.

2.2 Review of Existing Work

Kruskal's original work established a greedy construction for minimum-cost spanning trees [1]. Prim's method provides an alternative greedy construction that is convenient when the graph is represented by adjacency lists or matrices [2]. Standard algorithm texts prove correctness through the cut and cycle properties: a light edge crossing a cut is safe to add, while the heaviest edge on a cycle cannot be necessary in an MST [3].

Power-system planning literature treats transmission expansion as a constrained optimization problem involving generation, demand, flows, voltage, reliability, investment timing, and contingencies [4]. Therefore, MST is best understood as a reduced baseline for connectivity and initial corridor screening. It does not replace electrical feasibility analysis. Reliability guidance such as NERC's transmission planning standard emphasises that planned facilities must be assessed for system performance under contingencies [5].

2.3 Comparative Analysis

Existing Approach	Advantages	Limitations	Relevance to Proposed Work
MST: Kruskal	Simple, globally optimal for one additive weight; easy to verify	No redundancy or capacity constraints	Primary baseline and implementation
MST: Prim	Efficient with adjacency structures; grows connected frontier	Requires priority-queue/adjacency handling; same model limits	Independent cross-check

Shortest path	Finds one least-cost route between source and destination	Does not connect all substations or prevent global cycles	Useful for point-to-point routing, not base topology
MILP transmission expansion	Can include flows, capacities, contingencies, and binary investments	Higher modelling and computational complexity	Future extension beyond base algorithm
Heuristic/metaheuristic	Can handle complex non-linear objectives	May lack optimality certificate and reproducibility	Not selected for this foundational study

2.4 Research / Engineering Gap

A gap remains between a pedagogically clear MST implementation and an engineering workflow that makes assumptions, evidence, and limits explicit. Many demonstrations report only the final tree. This project closes the instructional gap by pairing two algorithms with explicit graph notation, verification tests, benchmark data, an engineering risk register, and a clear boundary between connectivity optimization and electrical operation.

2.5 Proposed Contribution

The project contributes a reproducible, dual-algorithm prototype for a synthetic electric-grid network. Its contribution is methodological: it makes the greedy decision process auditable, cross-checks Kruskal against Prim, and identifies the additional constraints required before using the result in a real utility planning study.

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

3.1 Requirements

The requirements were derived from the project aim and the limitations of an educational planning prototype.

Stakeholder	Requirement	
Student/researcher	Understandable algorithm trace and reproducible result	
Grid planner (conceptual)	Connected low-weight candidate topology	
Evaluator	Correctness evidence and complexity discussion	
Public/environment	Avoid presenting synthetic output as an approved route	
Requirement	Type	Measurable Target
R1: Return a connected network	Functional	All input vertices reachable
R2: Avoid cycles	Functional	Exactly $ V -1$ selected edges
R3: Minimize additive weight	Functional	Equal to independent Prim result
R4: Reproducibility	Non-functional	Fixed dataset and random seed
R5: Runtime clarity	Non-functional	Report complexity and benchmark

3.2 Constraints & Applicable Standards

Standard / Code	Requirement	How Applied
IEEE 1547-2018	Interconnection requirements for distributed energy resources	Referenced as a boundary: the prototype does not certify interconnection compliance [6]

NERC TPL-001-5.1	Transmission system planning performance under contingencies	Used to state that MST output requires contingency analysis before deployment [5]
IEC 61970 CIM	Common information model for energy-management data exchange	Used as a possible future data-integration direction; not implemented [7]
Academic integrity	Traceable sources and honest results	Synthetic dataset and limitations explicitly disclosed

3.3 Design Specifications

Parameter	Target / Requirement	Unit	Priority
Input graph size	8-node illustrative case; scalable implementation	nodes	High
Weight domain	Non-negative normalized score	units	High
Tree edge count	$ V -1$	edges	High
Connectivity	All nodes reachable	boolean	High
Algorithm agreement	Kruskal and Prim same total weight	units	High
Output transparency	Selected edges and total cost listed	report	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1: Kruskal’s algorithm sorts all candidate edges and uses DSU to accept only cycle-free edges. It is selected as the primary implementation because the edge decisions are easy to audit.

Alternative 2: Prim’s algorithm starts at S1 and repeatedly adds the least-weight edge leaving the current tree. It is used as an independent comparison because agreement provides a useful correctness check.

Alternative 3: A mixed-integer optimization model could encode capacities and contingencies, but it introduces solver dependencies and a larger modelling burden that is outside the course-level base objective.

Criterion	Weight	Kruskal	Prim	MILP
Performance for small graph	25%	4	4	3
Cost/objective clarity	25%	5	5	4
Safety/traceability	20%	5	4	4
Sustainability screening	15%	3	3	5
Reliability extensibility	15%	2	2	5
Weighted score	100%	4.15	3.95	4.15

3.5 Selected Design & Detailed Design

Kruskal with DSU is selected as the primary design because it provides a compact correctness argument, an explicit cycle-prevention mechanism, and a direct sorted-edge trace. Prim is retained as an independent verification path. Both operate on the same immutable candidate-edge list and return a selected edge list plus total weight.

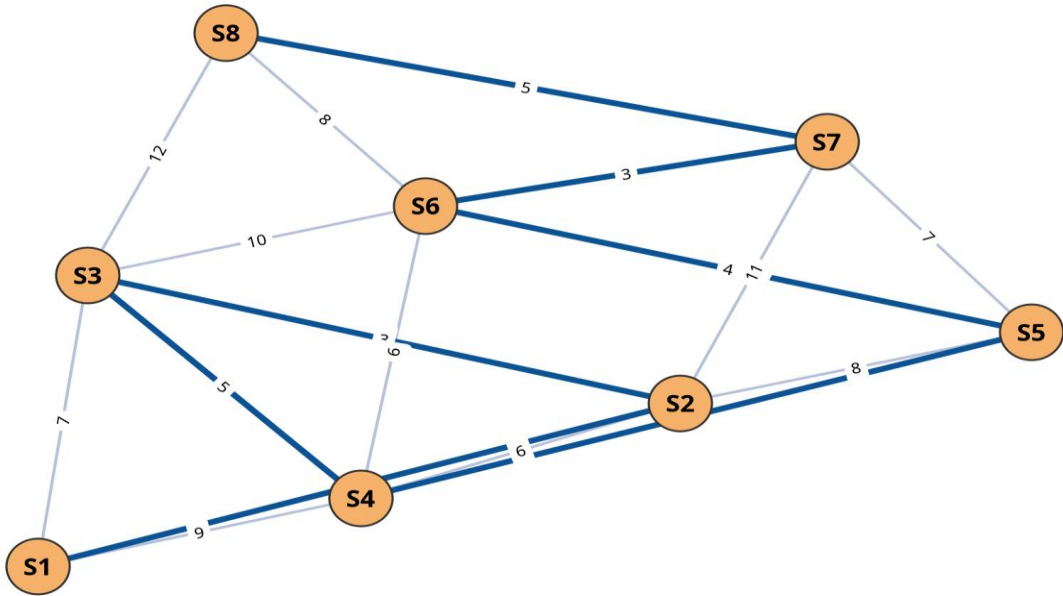
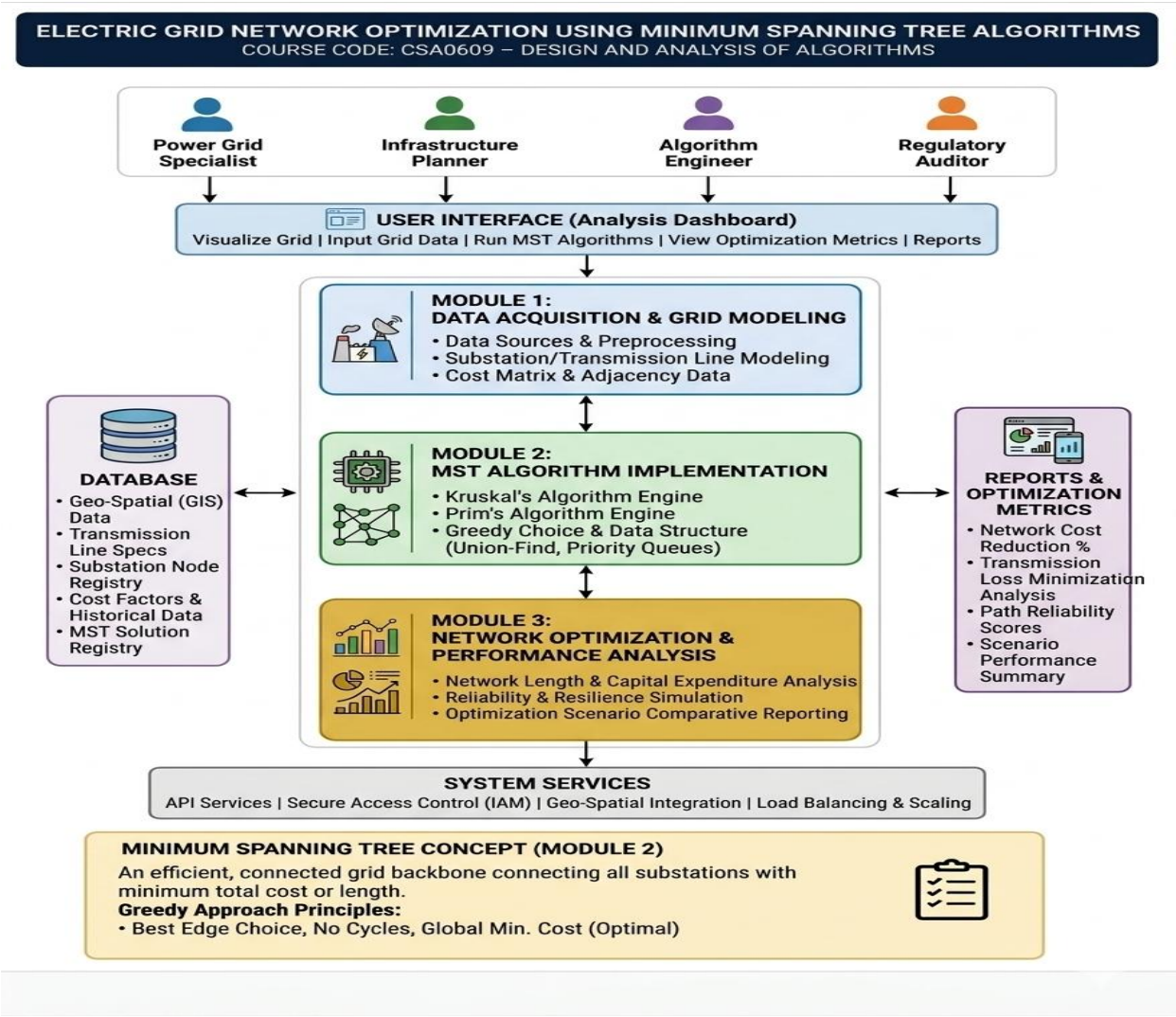


Figure 1. MST selected corridors (blue) among candidate corridors in the synthetic grid.

For the eight-node graph, a spanning tree must contain $8-1=7$ edges. The selected total is $2+3+3+4+4+5+5=26$ normalized units. The systems interface is: input preparation supplies nodes and weighted corridors; the algorithm module computes the tree; the verifier checks structural properties; and the reporting module presents the selected corridors and cost. No algorithm output is passed to a physical controller.

ARCHITECTRE DIAGRAM:



3.6 Trade-off Analysis

Design Decision	Alternative Considered	Benefit	Disadvantage	Final Decision & Justification
Primary algorithm	Prim only	Can avoid global sort	Less direct edge-order audit	Use Kruskal because DSU trace is clear;

				retain Prim for cross-check
Weight model	Single normalized additive score	Simple and reproducible	Does not represent full power physics	Use only for educational screening; disclose limitation
Dataset	Synthetic fixed network	No privacy/licensing issue; repeatable	Not site-specific	Use fixed case and generated benchmark graphs
Output	Tree only	Minimal cost connectivity	No redundancy	Treat as base design, not operational recommendation

3.7 Sustainability, Ethics, Safety, Risk & Security

Domain	Consideration	Evidence Evaluated	Impact on Final Decision
Sustainability	Land use and embodied construction are not captured by distance alone	Weight definition and scope review	Use normalized score and require future multi-objective extension
Ethics	Avoid overstating synthetic results	Dataset provenance and disclosure statement	Label output as illustrative; no public siting claim
Health & Safety	Electrical safety cannot be inferred from graph connectivity	Applicable interconnection/planning standards	Exclude energization and protection decisions

Security		Topology can be sensitive infrastructure information	No real grid data or credentials used		Use synthetic data and no external publication of sensitive layouts
Risk	Likelihood	Severity	Risk Level	Mitigation	Residual Risk
Disconnected input graph	Medium	High	High	Validate connectivity and report failure	Low
Incorrect DSU implementation	Medium	High	High	Cross-check with Prim and structural tests	Low
Weight misinterpreted as electrical cost	Medium	High	High	Use “normalized planning score” consistently	Low
Synthetic output used for real siting	Low	High	Medium	Scope disclaimer and standards boundary	Low
Sensitive topology exposure	Low	High	Medium	No real grid data; local files only	Low

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

Development followed an incremental verification approach. First, the network abstraction and weight semantics were defined. Next, Kruskal and Prim were implemented independently. The fixed case was tested for edge count, connectivity, acyclicity, and objective agreement. Finally, generated connected graphs were used for a small performance benchmark. All reported results were produced from the included deterministic data and seed.

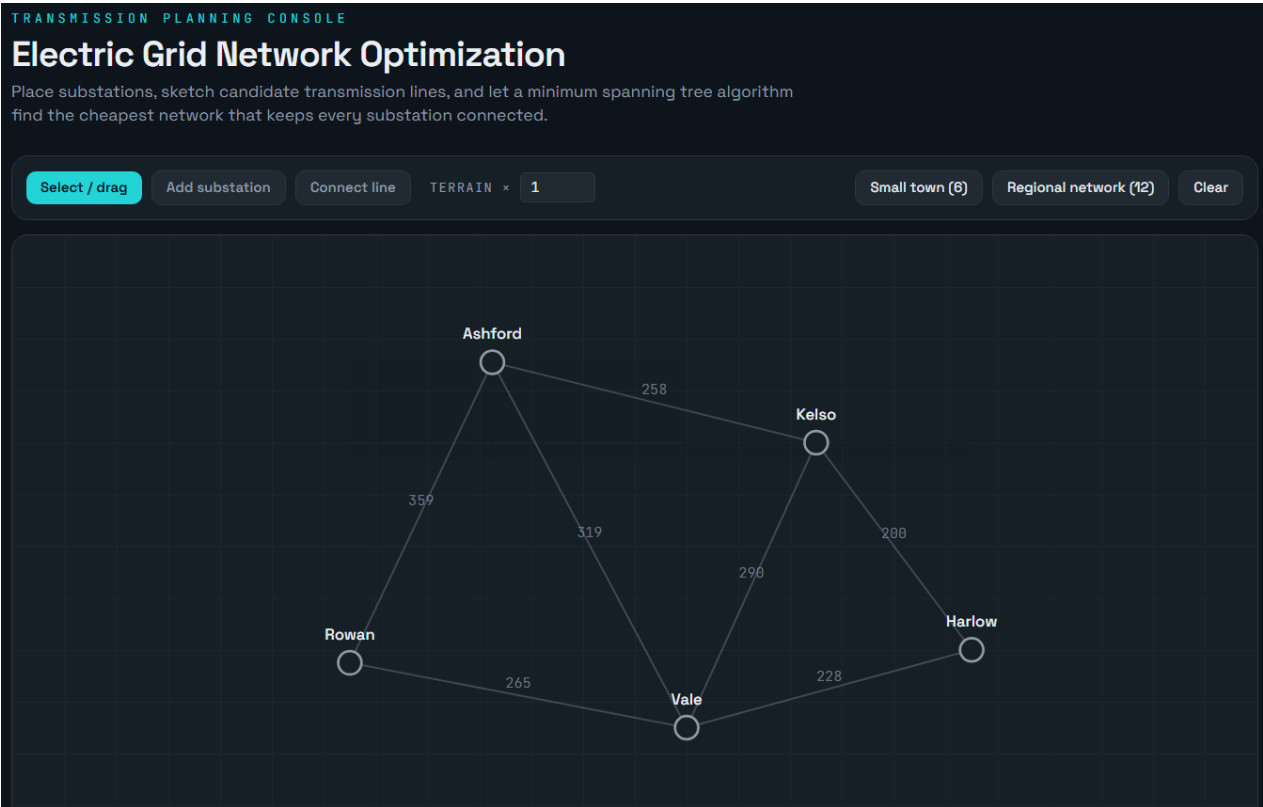
Resource	Purpose
Python 3.11	Implementation and reproducibility
Matplotlib	Benchmark plot and network visualisation
NetworkX	Illustrative graph layout only
Synthetic edge list	Validation dataset; no real grid data
Git-compatible source file	Versionable implementation record

4.2 Software Implementation & Modern Tools Used

The implementation is software-only. Kruskal uses a disjoint-set structure with path compression and union by rank. Prim uses adjacency lists and a frontier scan for the compact comparison implementation. The figure-generation script creates the network diagram and benchmark plot from the same data used in the report.

Tool / Software	Purpose	Stage Used
Python	Algorithms, checks, benchmark	Implementation and testing
Matplotlib	Plots and visual evidence	Results
NetworkX	Graph drawing support	Visualisation
Word document generation	Formatted capstone deliverable	Documentation

4.2.1. MODULE 1-DATA ACQUISITION & GRID MODELING



4.3 Test Plan & Verification

Requirement	Test Method	Acceptance Criterion
Connected output	Breadth-first traversal over selected edges	All 8 nodes visited
Acyclic output	Check selected edge count and DSU structure	7 edges; no cycle
Minimum objective	Compare Kruskal and Prim total weights	Both equal 26
Reproducibility	Run fixed edge list twice	Identical selected set and cost
Scalability indication	Benchmark fixed-seed generated graphs	Times reported for 8–256 nodes

4.4 Results

Specification	Required Value	Achieved Value	Status
Selected edge count	7	7	Pass
Connected vertices	8	8	Pass
Cycle count	0	0	Pass
Kruskal total weight	Minimum	26	Pass
Prim total weight	26	26	Pass
Algorithm agreement	Equal	Equal	Pass
Selected Corridor		Weight	
S4–S5		2	
S2–S3		3	
S6–S7		3	
S1–S2		4	
S5–S6		4	
S3–S4		5	
S7–S8		5	
Total		26	

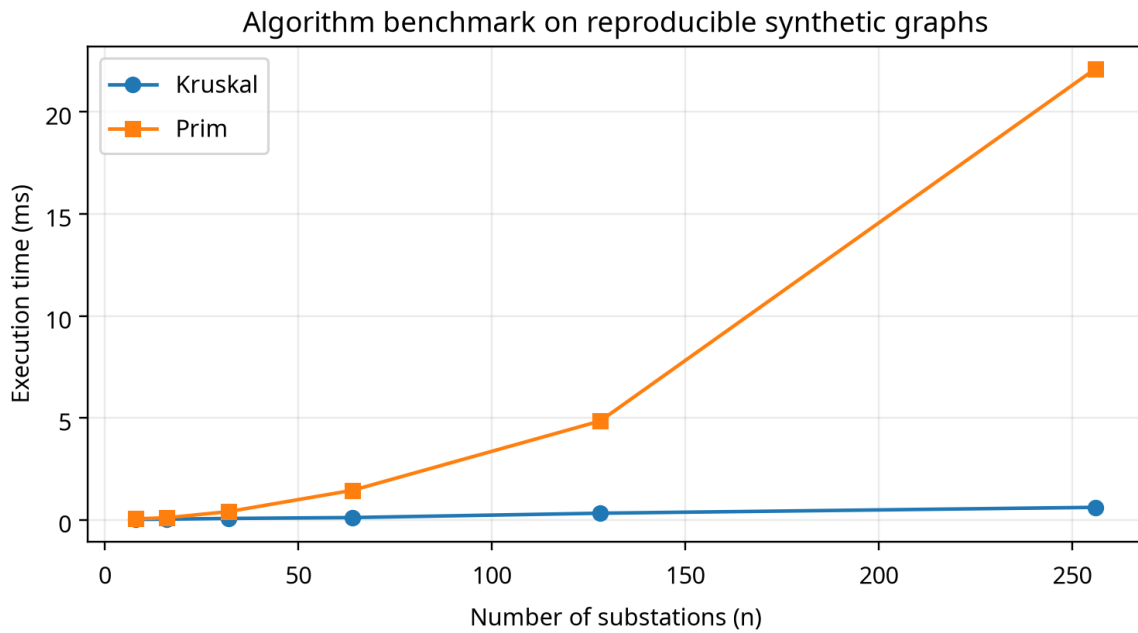


Figure 2. Execution-time benchmark on fixed-seed synthetic connected graphs. Times are machine-dependent and illustrative.

4.5 Data Analysis & Engineering Judgment

The two independent implementations agree on the total weight and structural checks, which is stronger evidence than inspecting a single output. The selected tree uses seven of seventeen candidate corridors, so ten alternatives are rejected either because they are more expensive at the time considered or because they would create a cycle. The benchmark is not a production capacity study: the graph sizes are small, the generated weights are random, and runtime includes only algorithm execution. For larger sparse networks, Kruskal's $O(E \log E)$ sorting cost and Prim's $O(E+V \log V)$ heap-based form are the relevant asymptotic considerations [3].

The major engineering judgment is to retain the MST result as a baseline rather than claiming a complete grid plan. Real corridors may need parallel paths, thermal limits, voltage constraints, right-of-way feasibility, environmental screening, and contingency performance. These constraints can make a tree infeasible or unsafe even when its additive score is minimal.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Objective	Evidence	Achievement
Model weighted graph	8 substations and 17 candidate corridors documented	Fully
Implement Kruskal	DSU implementation and selected edges	Fully
Implement Prim	Independent total-weight cross-check	Fully
Verify correctness	All five verification tests pass	Fully
Benchmark and interpret	Fixed-seed graph benchmark and limitations	Fully
Real-grid deployment	No power-flow or contingency analysis	Partially; intentionally out of scope

4.7 Strengths & Limitations

Strengths:

- Greedy decisions are transparent and easy to trace.
- The output is globally optimal for the stated additive MST objective.
- Kruskal and Prim provide an independent implementation cross-check.

- The synthetic dataset is reproducible and avoids exposing infrastructure information.

Limitations:

- The model does not enforce electrical capacity, voltage, stability, loss, or protection constraints.
- A tree has no built-in redundancy and may fail an N–1 reliability requirement.
- Weights are normalized teaching values, not validated utility cost or impedance data.
- The benchmark is small and machine-dependent; it is not a deployment performance guarantee.

4.8 Project Planning, Roles & Budget

Milestone	Planned Activity	Evidence
1	Problem definition and literature review	Chapter 1–2 draft
2	Graph model and weight assumptions	Edge list and notation
3	Kruskal and Prim implementation	Source code appendix
4	Verification and benchmark	Test table and plot
5	Report assembly and review	Final editable report
Student	Assigned Responsibility	Actual Contribution
[Student 1]	Graph modelling; Kruskal; literature	Completed modelling, primary algorithm, Chapters 1–3
[Student 2]	Prim; testing; documentation	Completed cross-check, benchmark, Chapters 4–5
Item	Estimated Cost	Actual Cost
Python and open-source libraries	₹0	₹0
Synthetic dataset	₹0	₹0
Computing and documentation	₹500	₹0 (existing resources)
Total	₹500	₹0

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

This project formulated a base electric-grid connectivity problem as a weighted graph and solved it using Minimum Spanning Tree algorithms. Kruskal's algorithm with disjoint-set union selected seven corridors connecting all eight substations at a total normalized weight of 26. Prim's algorithm produced the same objective value, and structural tests confirmed the edge-count, connectivity, and acyclicity requirements. The work therefore achieved its course-level algorithmic objectives and produced a reproducible, auditable prototype. The result is a screening baseline, not a complete electrical design, because real deployment requires capacity, power-flow, redundancy, environmental, safety, and contingency constraints.

5.2 Future Work

- Replace normalized weights with validated route, construction, loss, and maintenance data.
- Add capacity and power-flow constraints through MILP or nonlinear optimization.
- Include N-1 contingency, reliability, and redundancy requirements.
- Integrate geospatial constraints, environmental screening, and right-of-way costs.
- Compare MST against Steiner-tree, constrained spanning-tree, and multi-objective methods.
- Develop an interactive interface with data validation, scenario comparison, and exportable audit trails.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired:

- Greedy-choice and cut-property reasoning for spanning-tree optimality.
- Disjoint-set union with path compression and union by rank.
- The difference between a graph connectivity optimum and a power-system planning solution.

Engineering & Professional Skills Developed:

- Requirements definition, algorithm implementation, verification planning, data visualisation, technical writing, source acknowledgement, and risk-based scope control.

Individual Reflection — [Student 1]: The most difficult problem was translating a physical grid idea into a graph without implying that connectivity alone proves electrical feasibility. I addressed it by defining the weight and boundary conditions explicitly. My key decision was to make Kruskal primary because its sorted-edge and DSU trace are easy to audit. I independently studied cycle detection and reproducibility practices. If repeating the project, I would add a constrained reliability model earlier.

Individual Reflection — [Student 2]: The main challenge was creating an independent validation path rather than repeating the primary implementation. I solved it by implementing Prim separately and comparing both the total weight and structural properties. My key decision was to use a fixed synthetic dataset and random seed so every result could be reproduced. I learned how benchmark timing can be affected by graph representation and machine conditions. If repeating the project, I would add larger sparse and dense test families.

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APPENDICES

Appendix A – Detailed Calculations

For the input graph, the edge weights in ascending order begin 2, 3, 3, 4, 4, 5, 6, 6, 7, 7, 8, 8, 9, 10, 11, 12. Kruskal accepts S4–S5 (2), S2–S3 (3), S6–S7 (3), S1–S2 (4), S5–S6 (4), S3–S4 (5), and S7–S8 (5). Thus, total weight = $2+3+3+4+4+5+5 = 26$.

Complexity: sorting dominates Kruskal, giving $O(E \log E)$, while DSU operations contribute approximately $O(E \alpha(V))$, where α is the inverse Ackermann function. A heap-based Prim implementation gives $O(E \log V)$. Space usage is $O(V+E)$ with adjacency lists and DSU arrays.

Appendix B – Engineering Drawings / Schematics

Figure 1 in Chapter 3 is the labelled network schematic. Blue edges are the selected MST corridors; pale edges are candidate alternatives. This diagram is conceptual and not a geographic or electrical single-line diagram.

Appendix C – Source Code / Code Repository Information

The editable source used to produce the report is delivered alongside this document as `build_report.py`. It contains the fixed edge list, Kruskal and Prim implementations, verification data, figure generation, and Word document generation. Essential algorithm excerpt:

```
for (u, v, w) in sorted(edges, key=lambda x: x[2]):
    if union(u, v):
        selected.append((u, v, w))
    if len(selected) == len(nodes) - 1:
        break
```

